

Introducción a los Sistemas Operativos

Administración de Archivos - III



- ✓ Versión: Octubre 2012
- ✓ Palabras Claves: Archivo, File System, Directorio, UNIX, I-NODO, Windows, FAT

Algunas diapositivas han sido extraídas de las ofrecidas para docentes desde el libro de Stallings (Sistemas Operativos) y el de Silberschatz (Operating Systems Concepts). También se incluyen diapositivas cedidas por Microsoft S.A.



UNIX - Manejo de archivos

☑ Tipos de Archivos

- ✓ Regular, or ordinary
- ✓ Directory
- ✓ Special (representar dispositivos como archivos)
- ✓ Named pipes (comunicación entre procesos)
- ✓ Links (comparten el i-nodo)
- ✓ Symbolic links (para filesystems diferentes)



UNIX - Estructura del Volumen

- ✓ Boot Block: Código para bootear el S.O.
- ✓ Superblock: Atributos sobre el File System
- ✓ I-NODE Table: Tabla que contiene todos los I-NODOS
 - ✓ I-NODO: Estructura de control que contiene la información clave de un archivo
- ✓ Data Blocks: Bloques de datos de los archivos



UNIX - I-NODO

Table 12.4 Information in a UNIX Disk-Resident Inode

File Mode	16-bit flag that stores access and execution permissions associated with the file. 12-14 File type (regular, directory, character or block special, FIFO pipe) 9-11 Execution flags 8 Owner read permission 7 Owner write permission 6 Owner execute permission 5 Group read permission 4 Group write permission 3 Group execute permission 2 Other read permission 1 Other write permission 0 Other execute permission
Link Count	Number of directory references to this inode
Owner ID	Individual owner of file
Group ID	Group owner associated with this file
File Size	Number of bytes in file
File Addresses	39 bytes of address information
Last Accessed	Time of last file access
Last Modified	Time of last file modification
Inode Modified	Time of last inode modification



UNIX - I-NODO

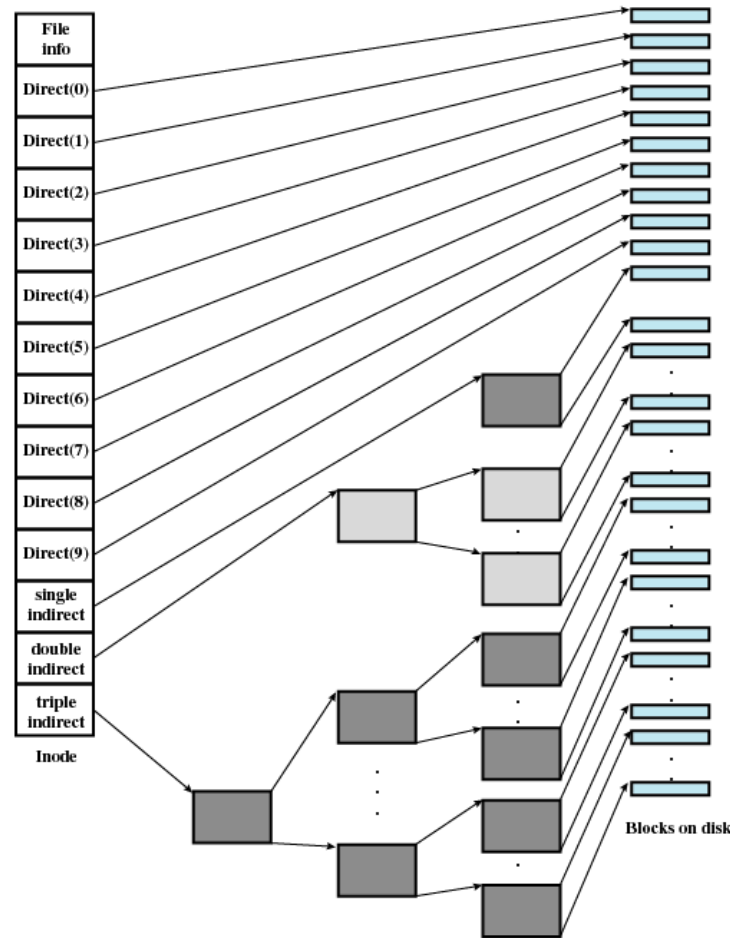
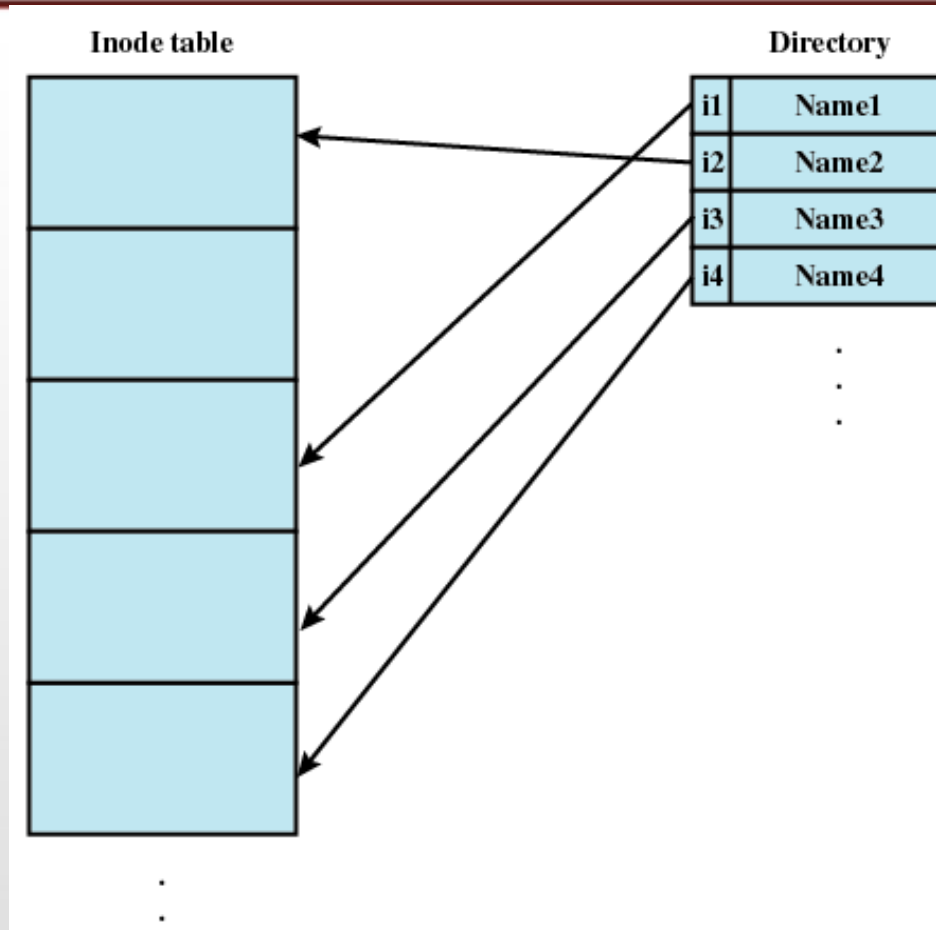


Figure 12.13 Layout of a UNIX File on Disk



UNIX - Directorios



UNIX - Directorios (cont)

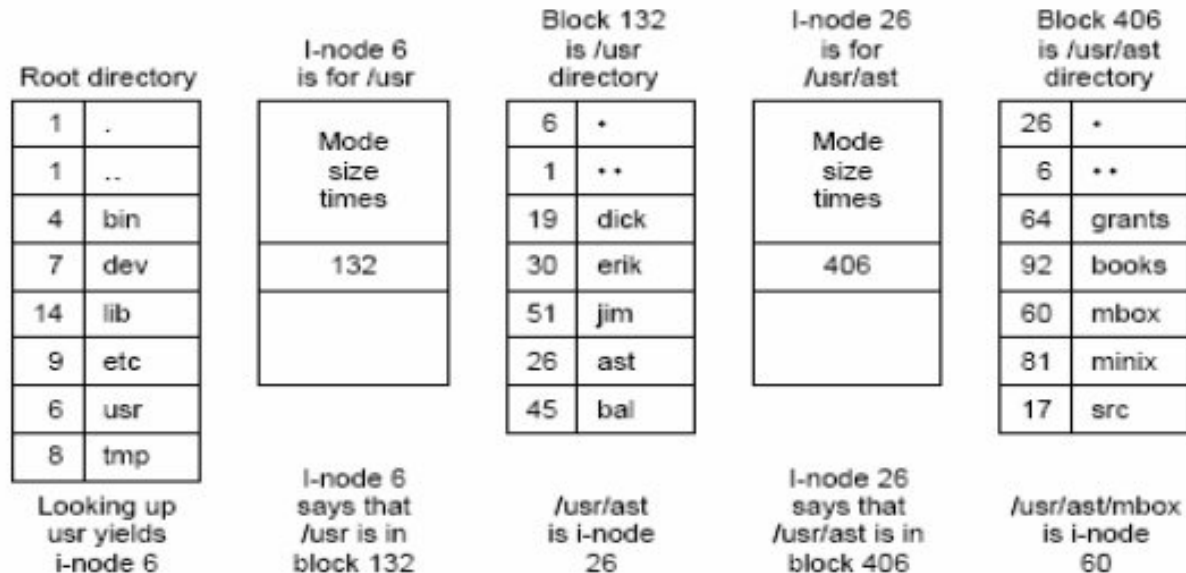


Figure 5-14. The steps in looking up `/usr/ast/mbox`.



LINUX - VFS - Virtual File System

- ✓ Interfase uniforme para el acceso de los procesos a los archivos
- ✓ Representa cualquier tipo de File System
- ✓ Asume que los archivos son objetos que comparten propiedades mas allá del File System que los almacena



LINUX - VFS (cont.)

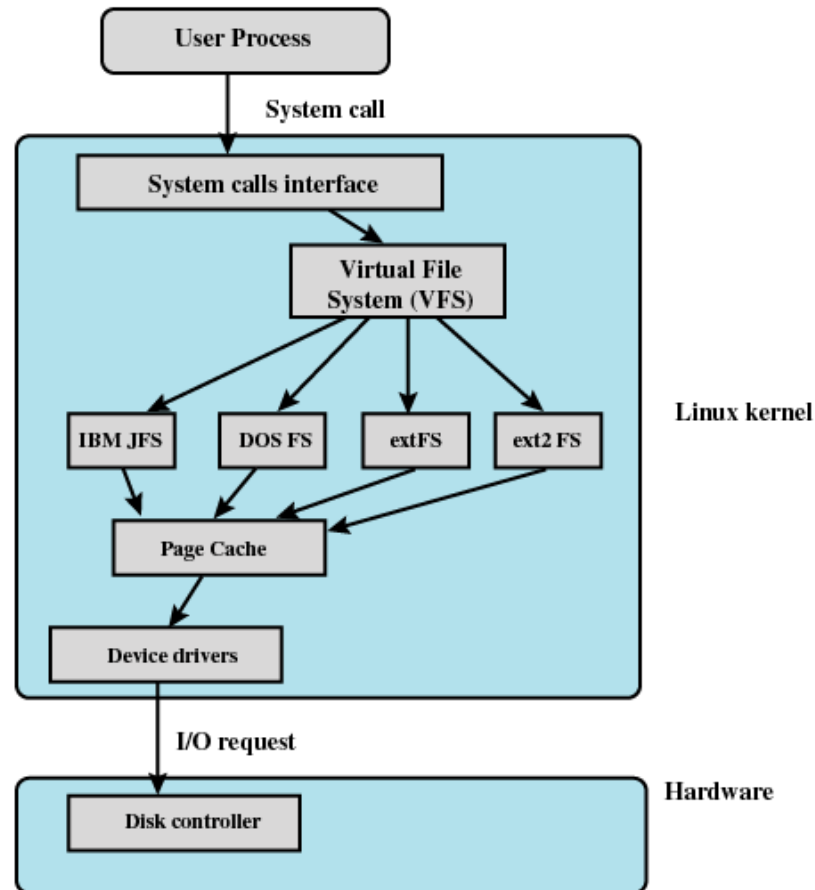
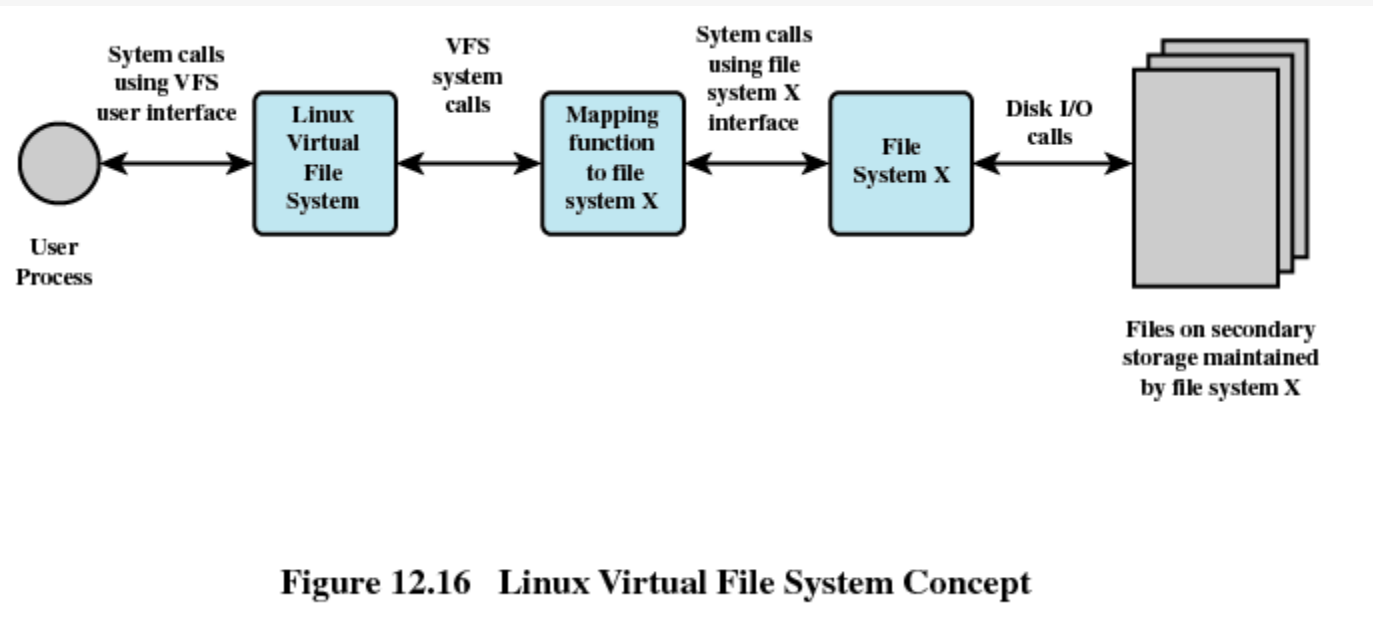


Figure 12.15 Linux Virtual File System Context



LINUX - VFS (cont.)



LINUX - VFS (cont.)

- ☑ Características similares a UNIX FileSystem
 - ✓ Estructura de Directorios Jerárquica
 - ✓ Directorios son tratados como archivos
 - ✓ I-NODOS son utilizados para describir los archivos



LINUX -VFS (cont.)

☑️ Objetos

- ✓ Superblock object
 - ♦ Representa el File System montado
- ✓ Inode object
 - ♦ Representa un archivo
- ✓ D-entry object
 - ♦ Representa una entrada en un directorio
- ✓ File object
 - ♦ Representa un archivo abierto asociado a un proceso



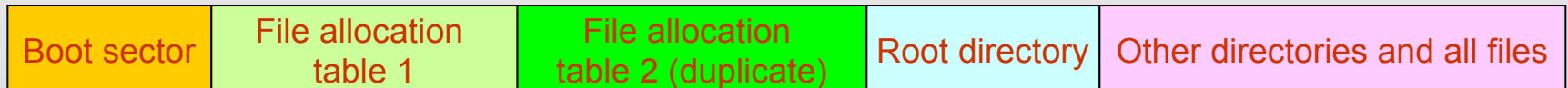
Windows - File Systems Soportados

- ☑ CD-ROM File System (CDFS)
- ☑ Universal Disk Format (UDF)
- ☑ File Allocation Table (FAT12, FAT16, and FAT32)
- ☑ New Technology File System (NTFS)



Windows - FAT

- ☑ FAT (File Allocation Table) file systems are a legacy format that originated in DOS and Windows 9x
- ☑ Reasons why Windows supports FAT file systems:
 - ✓ to enable upgrades from other versions of Windows
 - ✓ compatibility with other operating systems in multiboot systems
 - ✓ as a floppy disk format
- ☑ Each FAT format includes a number that indicates the number of bits the format uses to identify clusters on a disk



FAT format organization



Windows - FAT12

- ☑ FAT12's 12-bit cluster identifier limits a partition to storing a maximum of 2^{12} (4096) clusters
 - ✓ Windows uses cluster sizes from 512 bytes to 8 KB in size, which limits a FAT12 volume size to 32 MB
 - ✓ Windows uses FAT12 as the format for all 5-inch floppy disks and 3.5-inch floppy disks, which store up to 1.44 MB of data



Windows - FAT16

- ✓ FAT16, with a 16-bit cluster identifier, can address 2^{16} (65,536) clusters
 - ✓ On Windows, FAT16 cluster sizes range from 512 bytes (the sector size) to 64 KB, which limits FAT16 volume sizes to 4 GB
 - ✓ The cluster size Windows uses depends on the size of a volume



Windows - FAT32

- ✓ FAT32 is the most recently defined FAT-based file system format
- ✓ FAT32 uses 32-bit cluster identifiers but reserves the high 4 bits, so in effect it has 28-bit cluster identifiers
 - ✓ Because FAT32 cluster sizes can be as large as 32 KB, FAT32 has a theoretical ability to address 8 TB volumes
 - ✓ Although Windows works with existing FAT32 volumes of larger sizes (created in other operating systems), it limits new FAT32 volumes to a maximum of 32 GB
 - ✓ FAT32's higher potential cluster numbers let it more efficiently manage disks than FAT16; it can handle up to 128-MB volumes with 512-byte clusters



Windows - NTFS

- ☑ NTFS is the native file system format of Windows
- ☑ NTFS uses 64-bit cluster indexes
 - ✓ Theoretical ability to address volumes of up to 16 exabytes (16 billion GB)
 - ✓ Windows 2000 limits the size of an NTFS volume to that addressable with 32-bit clusters, which is 128 TB (using 64-KB clusters)
- ☑ Why use NTFS instead of FAT? FAT is simpler, making it faster for some operations, but NTFS supports:
 - ✓ Larger file sizes and disks
 - ✓ Better performance on large disks, large directories, and small files
 - ✓ Reliability
 - ✓ Security

